

RECOVERY FINE-TUNING RECOVERS WHERE IT WAS TRAINED: DURATION- AND DOMAIN-DEPENDENT GAINS IN PRUNED TEXT-TO-AUDIO DIFFUSION

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ABSTRACT

Structured pruning of text-to-audio diffusion models is followed by recovery fine-tuning, whose benefit is usually certified by one score at one inference setting. Using the released pruned and fine-tuned AudioLDM-M checkpoints at two pruning severities (65% and 83% of U-Net parameters removed), we measure the paired recovery gain in text–audio alignment across clip duration and prompt domain, anchored by a shuffled-caption chance floor and the real audio of the same prompts. At 83% pruning, fine-tuning closes 63% of the pruned checkpoint’s gap to real audio at its own 10.24s fine-tuning duration but only 33% at 3.84s, and nothing on held-out hip-hop captions; against the unpruned model, 82% versus 44% at 83% and 52% versus 8% at 65%. The duration dependence was pre-specified, replicates on a disjoint prompt set, survives a family-wise correction, is not a scorer artefact, grows monotonically over four durations, holds at the published sampler recipe, and is reproduced at both durations by a second scorer and by two event-level metrics outside the CLAP family. Recovery should be reported across operating points; lacking a matched dense fine-tuned control and human ratings, our claims concern evaluation, not mechanism.

Index Terms— text-to-audio generation, diffusion models, structured pruning, recovery fine-tuning, evaluation

1. INTRODUCTION

Structured pruning followed by recovery fine-tuning is the standard route to cheaper diffusion models [1–3], recently applied to text-to-audio (TTA) generation with AudioLDM [4, 5]. Recovery is judged by one aggregate number on the in-domain test set at one sampler setting, yet a generative model is used at an inference operating point: a clip duration, a prompt domain, a sampler configuration. Recovery fine-tuning is itself training at one operating point; nothing guarantees that its gain holds elsewhere.

We ask: does the gain of recovery fine-tuning over the pruned checkpoint depend on the operating point at which it is evaluated? We take the released pruned and fine-tuned AudioLDM-M-Full checkpoints of Singh et al. [5] at two pruning severities, train nothing, and measure the paired recovery gain in text–audio alignment across two clip durations and two prompt domains, read against a chance floor and a real-audio ceiling. Our findings:

- Duration. The recovery gain is several times larger at the fine-tuning duration (10.24s) than at 3.84s, at both

severities, and grows monotonically over four durations at severity 2. The interaction was pre-specified, replicates on a disjoint prompt set, and is not a scale or scorer effect (the chance floor moves by at most 0.025 between durations).

- Domain. At matched duration the gain is present in-domain (AudioCaps) and absent or negative on held-out hip-hop/rap captions—at both severities at 3.84s and at both durations at severity 2—where alignment after fine-tuning sits 0.02–0.03 above the chance floor, against 0.32 in-domain.
- Where the gain lives. Frame-level grounding shows the native gain spread uniformly over the clip; a crop analysis shows the short-duration deficit comes from generating short clips, not from scoring short excerpts.

2. BACKGROUND

Pruning and recovery in diffusion models. Structural pruning with fine-tuning recovery is established for image diffusion models [1–3] and self-supervised speech encoders [6, 7]. For TTA, Singh et al. [5] L1-prune the AudioLDM U-Net, fine-tune on AudioCaps for 10^6 steps and report in-domain FAD/KL at a fixed sampler; their fine-tuned pruned model even surpasses the unpruned one in-domain (FAD 1.57 vs. 3.95), a first sign of in-domain specialisation. All evaluate recovery at one operating point.

Evaluating text-to-audio generation. CLAP similarity [8] is the standard automatic proxy for text–audio alignment [9, 10]; FAD [11] is sensitive to sample size and embedding [12, 13]; event classifiers [14], human-aligned CLAP [15] and frame-level grounding [16] measure other axes.

Duration as an operating point. Latent audio diffusion conditions on the latent length and is trained at a fixed duration (10.24s for AudioLDM [4, 17]; newer systems condition on timing explicitly [18]), so duration is a natural operating point for a checkpoint fine-tuned at one; fine-tuning also trades in-distribution gains for out-of-distribution losses [19], motivating the domain axis.

3. EXPERIMENTAL DESIGN

3.1. Systems

The dense model is AudioLDM-M-Full [4] (U-Net stage multipliers (1, 2, 3, 5), 415.96 M U-Net parameters), itself

AudioCaps-fine-tuned by its authors. We study the two released pruning severities of [5]: severity 1, (1, 2, 3, 1), 145.67 M parameters (−65.0%), and severity 2, (1, 2, 1, 1), 71.08 M (−82.9%; 39% fewer multiply–accumulates [5]). At each severity we compare the pruned checkpoint (P) with the fine-tuned checkpoint (P+FT), the released result of 10^6 steps of AudioCaps recovery fine-tuning of P [5] (EMA weights). The released pruned checkpoints carry no usable EMA weights, so P is the released L1 channel selection applied to the dense EMA weights (bit-exact to the released raw weights at severity 1; at severity 2 the release differs in three decoder-seam tensors, so both conventions are evaluated: A’ primary, B’ sensitivity). “Recovery” names the fine-tuning stage, not an achieved restoration; we train no system. Two of the present authors released the evaluated checkpoints [5]; the operating points, batteries, estimands and gates below were specified independently.

3.2. Operating points and prompt batteries

Two durations: the native point (10.24 s, latent length 256; the fine-tuning duration) and a short point (3.84 s, latent length 96); a sweep adds 5.12 and 7.68 s (latent 128, 192) at severity 2. Two domains: in-domain AudioCaps test captions [20] and held-out hip-hop/rap captions from MusicCaps [21], a sub-domain with near-zero exposure in the recovery corpus (“music”; captions seven times longer, median 56.5 vs. 8 words, so the domain axis bundles content with caption style). Severity 2 uses 192 AudioCaps prompts (disjoint from all severity-1 prompts) and 64 music prompts (3 replicates at 3.84 s, one at 10.24 s); severity 1 uses 96 AudioCaps prompts for the pre-specified domain test, a matched 80-prompt subset for the duration comparison, and 64 music prompts at 3.84 s; the dense model is generated at both durations on both AudioCaps sets (matched controls, same noise). Generation uses DDIM [22], 50 steps, guidance 2.5, $\eta = 0$, fp32, single generation, EMA weights (off the published recipe, Sec. 5; a 64-prompt severity-2 check uses the published DDIM 200, guidance 3.5).

3.3. Paired protocol, scoring and anchors

Within each operating point the compared systems receive identical generation noise x_T per prompt (common random numbers); across durations the latent shapes differ, so duration contrasts are prompt-paired but not noise-paired. The primary scorer is fused CLAP (laion/clap-htsat-fused, rev. 365dea6e) [8], which repeat-pads 3.84 s audio to 10 s and centre-crops 10.24 s audio to 10 s; every (system, operating point) cell is scored with one fixed-order, fixed-seed call, part of the endpoint. Two anchors make absolute scores readable: a cell’s chance floor, the mean cosine between each clip and the captions of the other prompts in its battery (a shuffled-caption baseline from the same embeddings), and the real-audio ceiling, the real AudioCaps clip of each prompt, band-limited to 16 kHz and scored under the identical convention at full length and as its first 3.84 s. The unit is the prompt (music replicates averaged first); intervals are 95% prompt-level percentile bootstrap intervals ($B = 10^4$).

3.4. Quantities and analysis plan

Let R be the mean per-prompt recovery gain, $\text{CLAP}(P+FT) - \text{CLAP}(P)$, at a given (domain, duration); R_{nat} , R_{short} , R_{mus} denote AudioCaps-native, AudioCaps-short and music-short. The duration interaction is $J = R_{\text{nat}} - R_{\text{short}} = s(P+FT) - s(P)$, where the duration response $s(\cdot)$ is a system’s mean per-prompt score change from 3.84 to 10.24 s; J is paired per prompt, with pre-specified gate $\log_5(J) > 0$. The recovery ratio $\rho = R/(\text{ref} - P)$ is the fraction of the pruned checkpoint’s gap to a reference (dense, or the real audio of the same prompts) that fine-tuning closes. The domain contrast at matched duration is $R_{\text{short}} - R_{\text{mus}}$; the smallest effect size of interest is 0.025. “Pre-specified” means estimand, gate and prompt set were committed before any score was seen: the severity-1 domain test (the original hypothesis; failed), the severity-1 duration follow-up, the severity-2 replication with seam sensitivity (primary), the severity-2 music cell at 10.24 s and the secondary metrics. The FineLAP diagnostic, the severity-2 dense control, the duration sweep and the published-recipe check were registered after the primary result: committed, with their decision rules, before their own scores were seen, after the primary endpoint had been read. The severity-1 dense control, crop, rank-scale, Holm and anchor analyses are post-hoc.

4. RESULTS

4.1. The recovery gain grows with clip duration

At the more severe operating point, recovery fine-tuning buys several times more alignment at its own duration than at 3.84 s. On the disjoint 192-prompt set, with estimands frozen before scoring, the interaction is clearly resolved, $J = +0.159$ (Fig. 1, Table 1). Fine-tuning raises CLAP alignment from 0.055 to 0.299 at the native duration ($R_{\text{nat}} = +0.244$) but only from 0.015 to 0.100 at 3.84 s ($R_{\text{short}} = +0.085$; resolved, not absent); the pruned checkpoint gains just $s(P) = +0.040$ from the longer clip, the fine-tuned one $s(P+FT) = +0.200$. The seam convention does not matter (B’: $J = +0.161$).

The effect is not an artefact of scale. The chance floor is close to zero and nearly duration-independent (Table 2: −0.048 to +0.020 across the AudioCaps cells, moving by at most 0.025 between durations for any system), so the interaction survives chance correction: $J_c = +0.191$ [+0.162, +0.220]. Real audio of the same prompts scores 0.274 at 3.84 s and 0.440 at 10.24 s ($s(\text{real}) = +0.167$ [+0.150, +0.184]); against that ceiling fine-tuning closes $\rho = 63\%$ of the pruned checkpoint’s gap at the native duration but 33% at 3.84 s.

The matched dense control separates the systems’ duration response from the scorer’s. At severity 1, on the same 80 prompts and under the same convention, the dense model’s response $s(\text{dense}) = +0.150$ (0.202 → 0.352) equals the pruned checkpoint’s, $s(P) = +0.149$ (paired difference −0.001 [−0.056, +0.055]), while the fine-tuned checkpoint responds more, $s(P+FT) = +0.193$ (excess over dense +0.043 [−0.020, +0.109]). The recovery gain itself is small there ($R_{\text{short}} = +0.008$, $R_{\text{nat}} = +0.052$, $J = +0.044$), and part of the pruned checkpoint’s raw response is a rise of its own floor (+0.019; the fine-tuned checkpoint’s floor does not move), so on chance-corrected scores the severity-1 interaction is resolved, $J_c = +0.066$ [+0.020, +0.111]. The severity-2

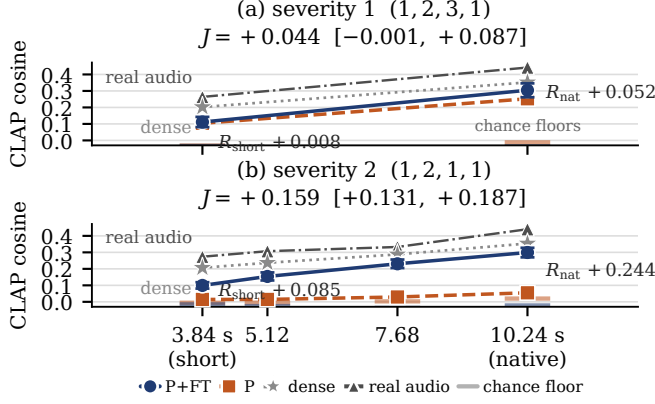


Fig. 1. The recovery gain grows with clip duration. Mean CLAP cosine of the pruned (P, dashed) and fine-tuned (P+FT, solid) checkpoints at the short and native points, (a) severity 1, and (b) severity 2 with the two sweep durations; whiskers: 95% CI of the paired gain R about the P+FT mean. Anchors: real audio of the same prompts (triangles), each cell’s chance floor (ticks) and the matched dense control (stars; same prompts and scoring convention).

control reads the same on its own prompts ($0.207 \rightarrow 0.354$): $s(\text{dense}) = +0.147 [+0.117, +0.177]$, against which the pruned checkpoint responds $-0.107 [-0.137, -0.076]$ less and the fine-tuned checkpoint $+0.053 [+0.017, +0.089]$ more (paired).

The gain grows monotonically with duration. A sweep of P, P+FT and dense over 5.12 and 7.68s on the same prompts (Fig. 1b) gives $R = +0.085, +0.139, +0.201, +0.244$ at 3.84, 5.12, 7.68, 10.24s, every step resolved ($+0.054 [+0.032, +0.077]$, $+0.062 [+0.037, +0.087]$, $+0.043 [+0.014, +0.073]$): by the pre-specified shape rule the gain is monotone increasing. The pruned checkpoint is flat to 5.12s (-0.000) while P+FT rises at every step; ρ_{dense} climbs 44%, 63%, 78%, 82% (Table 2); ρ_{real} dips at 10.24s only because the real clip itself gains $+0.108$ once it fills the scorer’s 10s window without repeat-padding.

Severity 1 is underpowered, not negative: its raw-scale interval narrowly includes zero (the follow-up was powered for $J \geq 0.065$ at $n = 80$), but it is directionally consistent on every scale (post-hoc): W rises from 0.44 to 0.64, the median per-prompt interaction is $+0.051 [+0.012, +0.077]$ and the second scorer gives $J_{\text{HC}} = +0.075 [+0.012, +0.137]$. A Holm correction over all 19 contrasts, sweep steps and published-recipe check included, leaves every severity-2 conclusion unchanged ($p < 10^{-4}$; last sweep step $p = 0.003$); at severity 1, R_{nat} ($p = 0.016$) and J ($p = 0.052$) do not survive, so the severity-1 duration effect rests on the severity-2 replication.

4.2. At matched duration the gain is domain-specific

The gain does not transfer off the fine-tuning domain, at either duration. At severity 2 and 3.84s it is resolved in-domain ($R_{\text{short}} = +0.085$) and null on music ($R_{\text{mus}} = +0.009$; $W = 0.48$), a matched-duration domain contrast of $+0.076$; at the native duration the music contrast is likewise null ($+0.005$; $W = 0.53$), so the large native gain is confined to the fine-tuning domain (domain contrast $+0.239$; music duration interaction $-0.004 [-0.043, +0.034]$). At severity 1 the pre-

Table 1. Recovery gain in CLAP alignment: mean cosine per checkpoint, paired gain R with 95% CI, and fraction W of prompts on which P+FT beats P. Duration rows give $s(P)$, $s(P+FT)$ and, in the R column, J (domain contrasts two-sample; n as in Sec. 3.2). †: CI excludes 0; ‡: gate passes.

Setting	P	P+FT	R [95% CI]	W
Severity 1, (1, 2, 3, 1)				
AudioCaps 3.84 s	0.104	0.111	$+0.008 [-0.023, +0.039]$	0.44
AudioCaps 10.24 s	0.253	0.304	$+0.052^\dagger [+0.009, +0.093]$	0.64
music 3.84 s	0.117	0.023	$-0.094^\dagger [-0.124, -0.065]$	0.20
duration $s; J$	$+0.149$	$+0.193$	$+0.044 [-0.001, +0.087]$	
domain 3.84 s ($n=96$)			$+0.092^\dagger [+0.054, +0.131]$	
Severity 2, (1, 2, 1, 1)				
AudioCaps 3.84 s	0.015	0.100	$+0.085^\dagger [+0.066, +0.105]$	0.72
AudioCaps 5.12 s	0.015	0.154	$+0.139^\dagger [+0.115, +0.164]$	0.79
AudioCaps 7.68 s	0.029	0.231	$+0.201^\dagger [+0.175, +0.227]$	0.89
AudioCaps 10.24 s	0.055	0.299	$+0.244^\dagger [+0.215, +0.273]$	0.87
music 3.84 s	0.005	0.014	$+0.009 [-0.013, +0.032]$	0.48
music 10.24 s	0.089	0.094	$+0.005 [-0.028, +0.039]$	0.53
duration $s; J$	$+0.040$	$+0.200$	$+0.159^\dagger [+0.131, +0.187]$	
domain 3.84 s			$+0.076^\dagger [+0.047, +0.105]$	
domain 10.24 s			$+0.239^\dagger [+0.195, +0.283]$	

Table 2. Anchors and recovery ratios (AC = AudioCaps; n as in Table 1): chance floor of the P/P+FT cells, real audio of the same prompts, and the fraction ρ of the pruned checkpoint’s gap to dense and to real audio closed by fine-tuning (95% CI, %). Sweep rows and severity-2 dense control registered after the primary result; severity-1 dense control post-hoc.

Setting	floor P/P+FT	real	ρ_{dense} [%]	ρ_{real} [%]	
sev. 1 AC 3.84 s	-0.031/-	-0.033	0.264	8% [-30, 36]	5% [-16, 24]
sev. 1 AC 10.24 s	-0.012/-	-0.036	0.442	52% [11, 103]	27% [6, 46]
sev. 2 AC 3.84 s	-0.005/-	-0.015	0.274	44% [36, 53]	33% [26, 39]
sev. 2 AC 5.12 s	-0.005/-	-0.025	0.307	63% [52, 74]	48% [40, 55]
sev. 2 AC 7.68 s	+0.003/-	-0.037	0.333	78% [68, 88]	66% [59, 74]
sev. 2 AC 10.24 s	+0.020/-	-0.022	0.440	82% [72, 92]	63% [56, 71]
sev. 1 music 3.84 s	+0.055/+	0.001	-	-	-
sev. 2 music 3.84 s	-0.013/-	0.004	-	-	-
sev. 2 music 10.24 s	+0.070/+	0.061	-	-	-

specified domain test (96 prompts) found no in-domain gain at 3.84s ($R_{\text{AC}} = -0.002 [-0.027, +0.021]$) and a negative music contrast ($R_{\text{mus}} = -0.094$; domain contrast $+0.092$).

The chance floor reads the two severities as one statement (Table 2): after fine-tuning, alignment on the hip-hop captions is 0.022, 0.018 and 0.033 above chance (severity 1 at 3.84s; severity 2 at 3.84 and 10.24s), against 0.115 and 0.321 in-domain at severity 2. Whether that registers as a “penalty” depends on whether the pruned checkpoint had music alignment to lose: the 65%-pruned one did (0.061 above chance; chance-corrected $R_{\text{mus}} = -0.040 [-0.060, -0.020]$), the 83%-pruned one did not (0.018). The null is near chance for both checkpoints (no real hip-hop reference was available for a ceiling) and not a caption-length effect: within AudioCaps the native gain is uncorrelated with caption length (Spearman $\rho = +0.04 [-0.12, +0.18]$), and no caption is truncated by the conditioner—though 47% of the music captions exceed CLAP’s 77-token pre-training length, a caption-style covariate inseparable from content.

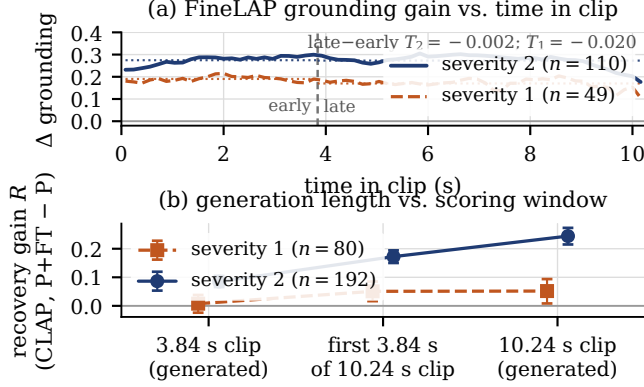


Fig. 2. (a) FineLAP frame-level grounding gain ($P+FT - P$) vs. time in the 10.24s clip : uniform, not back-loaded. (b) R (95% CI) on the generated 3.84s clip, the first 3.84s of the 10.24s clip and the full clip: a generation-length, not scoring-window, effect.

4.3. Where in the clip the gain lives

Two analyses locate the native-duration gain. First, a prospectively frozen diagnostic scores the native clips with FineLAP [16], a frame-level language-audio grounding model (EAT encoder; not CLAP-derived) reporting per 0.16s frame how strongly the requested event is grounded (110/49 eligible prompts at severities 2/1, outcome-blind). Fine-tuning raises grounding across the whole clip (severity 2: semantic mass $+0.27$ [$+0.22$, $+0.33$]; both severities; seam-robust), and the gain in the first 3.84s equals the gain in the remainder: $D_{\text{early}} = +0.275$ vs. $D_{\text{late}} = +0.273$, $T = -0.002$ [-0.024 , $+0.020$] (Fig. 2a). Second, a post-hoc crop analysis scores the first 3.84s of each 10.24s generation under the same convention against the separately generated 3.84s clips (Fig. 2b). The crop carries a large part of the native gain: at severity 1, $R_{\text{crop}} = +0.051$ [$+0.016$, $+0.085$] equals R_{nat} ($R_{\text{nat}} - R_{\text{crop}} = +0.001$ [-0.027 , $+0.029$]) and exceeds R_{short} by $+0.043$ [$+0.002$, $+0.085$]; at severity 2, $R_{\text{crop}} = +0.172$ [$+0.150$, $+0.194$], $R_{\text{crop}} - R_{\text{short}} = +0.087$ [$+0.065$, $+0.110$], $R_{\text{nat}} - R_{\text{crop}} = +0.072$ (seam-robust): the deficit is a generation effect, not a scoring one.

4.4. Corroboration and pre-specified negatives

At the severity-2 native point Human-CLAP [15] (a CLAP model fine-tuned on human ratings; automatic) gives $R_{\text{nat}} = +0.375$ [$+0.340$, $+0.408$] and $J = +0.185$; two pre-specified event-level metrics favour P+FT: KL to real references, 2.23 vs. 4.45 ($\Delta = +2.22$ [$+1.93$, $+2.53$]); PANNs top-10 captured labels, 1.46 vs. 0.60 ($\Delta = +0.86$ [$+0.70$, $+1.02$]). Rescored at 3.84s against the same references cropped alike (post-hoc), both gains shrink—KL $+0.66$ [$+0.42$, $+0.92$], PANNs $+0.19$ [$+0.06$, $+0.32$ —so the interaction holds outside the CLAP family: $J_{\text{KL}} = +1.56$ [$+1.19$, $+1.92$], $J_{\text{PANN}} = +0.67$ [$+0.49$, $+0.86$] (seam-robust). At the published sampler recipe (DDIM 200, guidance 3.5; single generation) on the first 64 prompts the interaction holds, $J = +0.184$ [$+0.126$, $+0.243$] (pre-specified gate; the frozen recipe on the same prompts gives $+0.168$, difference $+0.016$ [-0.040 , $+0.072$]). Off the primary scorer

(post-hoc), Human-CLAP, KL and PANNs capture all show R rising at every step of the sweep, the first two steps resolved and the last (7.68 $\rightarrow$ 10.24s) not: Human-CLAP is flat there ($+0.004$ [-0.033 , $+0.041$]), KL $+0.27$ [-0.04 , $+0.57$], PANNs $+0.10$ [-0.07 , $+0.27$].

Three pre-specified hypotheses failed and are kept: (i) that fine-tuning trades in-domain for out-of-domain alignment: it requires an in-domain gain at 3.84s, which severity 1 lacks ($R_{\text{AC}} \approx 0$), so the music loss is real but the trade is not; (ii) the severity-1 music penalty did not replicate (Human-CLAP alone shows -0.037 [-0.068 , -0.005] at severity 2); (iii) a “late allocation” account of the duration effect is rejected by FineLAP ($T \approx 0$). Finally, P+FT is not restored to dense: at severity 1 and 10.24s the dense model leads P by $+0.099$ [$+0.058$, $+0.140$] and P+FT by $+0.048$ [-0.000 , $+0.096$]; at severity 2 by $+0.055$ [$+0.021$, $+0.088$] at 10.24s ($+0.107$ at 3.84s).

5. DISCUSSION AND LIMITATIONS

Across severities, scorers and metrics, the recovery gain of the released fine-tuned AudioLDM-M checkpoints tracks the operating point of the fine-tuning itself: growing monotonically toward the fine-tuning duration, confined to the fine-tuning domain, heterogeneous across prompts. At 65% pruning the pruned checkpoint still responds to duration like the dense model and fine-tuning adds a modest in-domain increment. At 83% pruning it is barely above chance at either duration (0.020 and 0.035 above the floor), and fine-tuning restores a duration response above the dense model’s—only in the fine-tuning domain and increasingly toward the fine-tuning duration, consistent with the recovery stage acting in part as further in-domain, native-duration specialisation. Recovery studies should report a training-matched and an off-native operating point, absolute scores with a chance floor, paired gains with prompt-level uncertainty, and their interaction.

Limitations. One case study (one model family, two severities, two domains). Mechanistic attribution is blocked: the matched control (the dense model given the same AudioCaps fine-tune) no longer exists and the public dense text-fine-tuned release is not equivalent, so the dependence cannot be attributed to pruning rather than to fine-tuning in general. Primary inference rests on CLAP-family scorers; there is no human evaluation. Sampler settings differ from the published recipe (DDIM 50 vs. 200, guidance 2.5 vs. 3.5); the interaction is reproduced at DDIM 200/3.5 on a 64-prompt subset, best-of-3 selection was not. The sweep stops at the fine-tuning duration, so it cannot separate “largest at the training duration” from “larger for longer clips”. One held-out sub-genre (no real hip-hop ceiling); cross-severity comparisons are descriptive; the severity-1 duration test was underpowered.

6. CONCLUSION

Recovery fine-tuning of the released pruned AudioLDM-M checkpoints recovers much at its own operating point and much less away from it: at 83% pruning it closes 63% of the gap to real audio at the fine-tuning duration, 33% at 3.84s and none on held-out music. One restored score at one operating point does not describe that. Audio: gbbibo. github.io/audioldm-modality-swap-pruning.

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